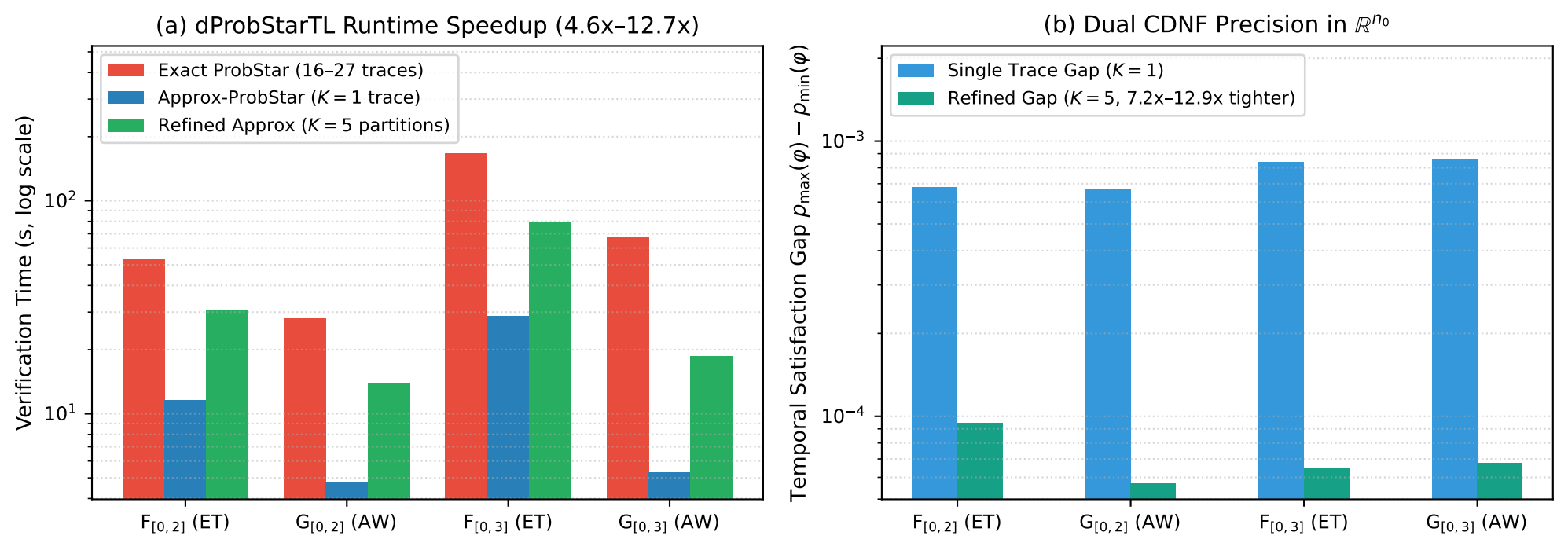
(a) dProbStarTL Runtime Speedup (4.6x–12.7x)
Exact ProbStar (16–27 traces)
Approx-ProbStar ($K = 1$ trace)
Refined Approx ($K = 5$ partitions)
Verification Time (s, log scale)
10^2
10^1
$F_{[0,2]}$ (ET)
$G_{[0,2]}$ (AW)
$F_{[0,3]}$ (ET)
$G_{[0,3]}$ (AW)
(b) Dual CDNF Precision in $\mathbb{R}^{n_0}$
Single Trace Gap ($K = 1$)
Refined Gap ($K = 5$, 7.2x–12.9x tighter)
Temporal Satisfaction Gap $p_{\max}(\varphi) - p_{\min}(\varphi)$
10^{-3}
10^{-4}
$F_{[0,2]}$ (ET)
$G_{[0,2]}$ (AW)
$F_{[0,3]}$ (ET)
$G_{[0,3]}$ (AW)